Mathematics for Social Scientists

Part 3
Math Applications in the Social Sciences
Linear Regression

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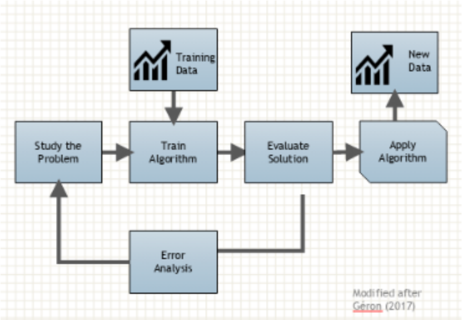
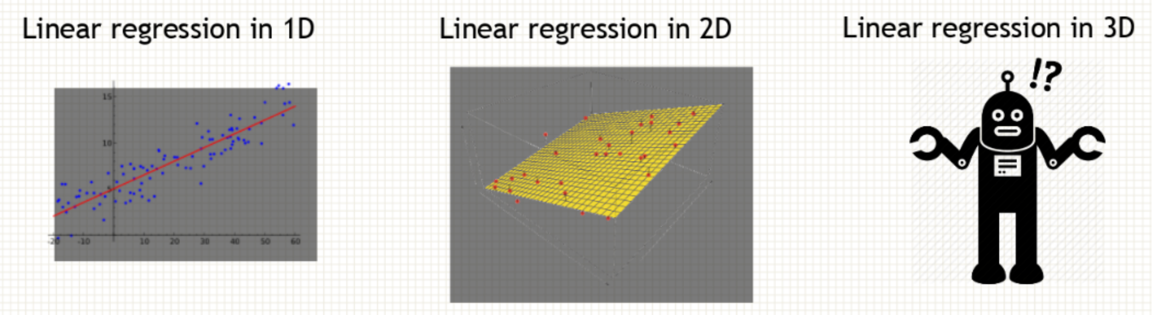
The problem of supervised learning

- Let x be a measurable quantity (input)
- Let y be a desirable quantity (output)
- Choose a family of models $\hat{y} = f(x; \theta)$ with free parameters θ



Linear Regression Illustration



Classical OLS assumptions

- Linearity in parameters: $y = X\beta + u$.
- $\mathbb{E}[u \mid X] = 0$.
- $\operatorname{rank}(X) = k$ (no perfect multicollinearity).
- $\mathsf{Var}(u \mid X) = \sigma^2\mathbf{I}$ and no serial correlation (for simple SEs).
- Normality of u (only for exact small-sample inference).

Parameter Estimation

Two approaches to minimizing the loss function:

- **The analytic approach**: obtain “best” parameters by finding a closed-form mathematical expression
- **The iterative approach**: Randomly initialize parameters. Update parameters in a stepwise manner via “informed guesses” (i.e. minimize the loss function in steps)

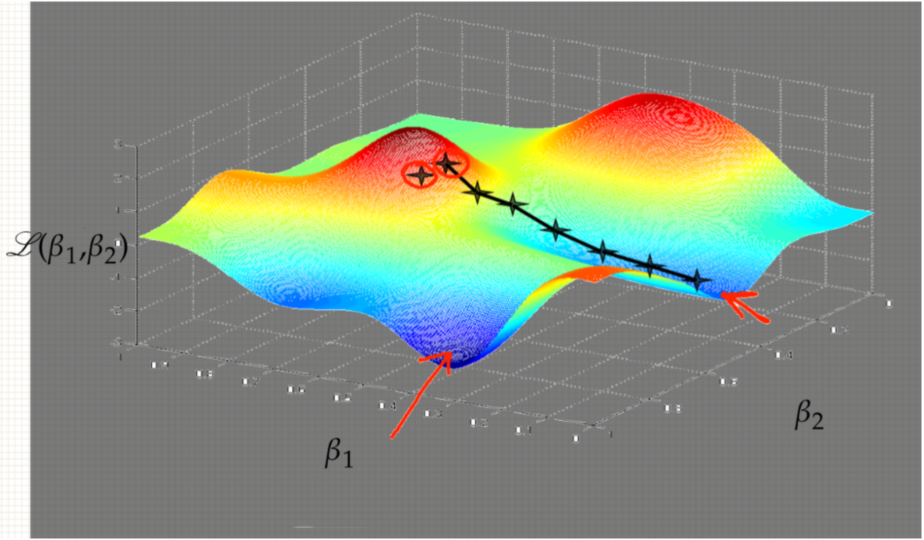
Parameter Estimation

- Usually, the software implementation takes care of parameter estimation. You only need to formulate the model.
- However, it is beneficial to (intuitively) understand what happens under the hood, and potentially detect meaningless results/errors in estimation.
- Let's illustrate the two approaches applied to linear regression!

The iterative approach

- Whenever an analytic solution is available, we can guarantee an optimal solution given the data
- If there is no analytic solution (or it is too costly to compute), we resort to stepwise algorithms which are guaranteed to approximate the optimal solution (and even converge to it, under certain conditions).
- The most commonly used iterative algorithm statistics and machine learning is **gradient descent**.

Gradient descent



The analytic approach

- Sometimes, it is (relatively) easy to find a closed form solution for the minimizer of the loss function.
- For linear regression, we use the fact that the MSE is a quadratic function of the parameters and thus has a single minimum. The minimizer is given by: $\hat{\beta} = (X'X)^{-1}X'y$.

Let's start with the scalar notations and build our way to the estimator above.

Simple Linear Regression (Scalar Form)

For observations indexed by $i = 1, \ldots, n$, consider the model

$$Y_i \;=\; \beta_0 + \beta_1 X_i + u_i.$$

- Y_i is the dependent variable; X_i is the independent variable; u_i is the disturbance.
- The least squares estimates $(\hat{\beta}_0, \hat{\beta}_1)$ minimize the sum of squared residuals

$$S(\beta_0, \beta_1) \;=\; \sum_{i=1}^{n} \left(Y_i - \beta_0 - \beta_1 X_i\right)^2.$$

- First-order conditions:

$$\sum_{i=1}^{n} (Y_i - \beta_0 - \beta_1 X_i) = 0, \qquad \sum_{i=1}^{n} X_i \,(Y_i - \beta_0 - \beta_1 X_i) = 0.$$

- Closed-form solution:

$$\hat{\beta}_1 = \frac{\sum_{i=1}^{n} (X_i - \bar{X})(Y_i - \bar{Y})}{\sum_{i=1}^{n} (X_i - \bar{X})^2}, \qquad \hat{\beta}_0 = \bar{Y} - \hat{\beta}_1 \bar{X}.$$

Multiple Linear Regression (Scalar Form)

With k regressors, the model for each observation $i = 1, \ldots, n$ is

$$Y_i \;=\; \beta_0 + \beta_1 X_{i1} + \beta_2 X_{i2} + \cdots + \beta_k X_{ik} + u_i.$$

- The least squares estimates $\hat{\beta}_0, \ldots, \hat{\beta}_k$ minimize

$$S(\beta_0, \ldots, \beta_k) \;=\; \sum_{i=1}^{n} \left(Y_i - \beta_0 - \beta_1 X_{i1} - \cdots - \beta_k X_{ik} \right)^2.$$

- First-order conditions (one equation per parameter):

$$\sum_{i=1}^{n} \left(Y_i - \beta_0 - \beta_1 X_{i1} - \cdots - \beta_k X_{ik} \right) = 0,$$

$$\sum_{i=1}^{n} X_{ij} \left(Y_i - \beta_0 - \beta_1 X_{i1} - \cdots - \beta_k X_{ik} \right) = 0, \quad j = 1, \ldots, k.$$

- These $k + 1$ linear equations motivate the *Matrix Form*, where the system is written compactly and solved using linear algebra.

The True Model (Matrix Form)

Our statistical model will essentially look something like the following:

$$\underbrace{\begin{bmatrix} Y_1 \\ Y_2 \\ \vdots \\ Y_n \end{bmatrix}}_{n \times 1} = \underbrace{\begin{bmatrix} 1 & X_{11} & X_{21} & \cdots & X_{k1} \\ 1 & X_{12} & X_{22} & \cdots & X_{k2} \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 1 & X_{1n} & X_{2n} & \cdots & X_{kn} \end{bmatrix}}_{n \times (k+1)} \underbrace{\begin{bmatrix} \beta_0 \\ \beta_1 \\ \beta_2 \\ \vdots \\ \beta_k \end{bmatrix}}_{(k+1) \times 1} + \underbrace{\begin{bmatrix} \varepsilon_1 \\ \varepsilon_2 \\ \vdots \\ \varepsilon_n \end{bmatrix}}_{n \times 1}$$

The True Model (Matrix Form)

This can be rewritten more simply as:

$$y = X\beta + \varepsilon \tag{1}$$

This is assumed to be an accurate reflection of the real world.
The model has a *systematic component* $(X\beta)$ and a *stochastic component* (ε).
Our goal is to obtain estimates of the population parameters in the vector β.

Criteria for Estimates

Our *estimates* of the population parameters are denoted by $\hat{\beta}$. The criterion we use is to find the estimator $\hat{\beta}$ that minimizes the *sum of squared residuals*.

The vector of residuals is

$$e = y - X\hat{\beta}. \tag{2}$$

Note: Carefully distinguish between *disturbances* ε (unobserved) and *residuals* e (observed). In general, $\varepsilon \neq e$.

Sum of Squared Residuals

The sum of squared residuals (RSS) is $e'e$.

$$\underbrace{\begin{bmatrix} e_1 & e_2 & \cdots & e_n \end{bmatrix}}_{1\times n} \underbrace{\begin{bmatrix} e_1 \\ e_2 \\ \vdots \\ e_n \end{bmatrix}}_{n\times 1} = \underbrace{\begin{bmatrix} e_1^2 + e_2^2 + \cdots + e_n^2 \end{bmatrix}}_{1\times 1}.$$

It follows that

$$\begin{aligned} e'e &= (y - X\hat{\beta})'(y - X\hat{\beta}) && (4) \\ &= y'y - \hat{\beta}'X'y - y'X\hat{\beta} + \hat{\beta}'X'X\hat{\beta} \\ &= y'y - 2\,\hat{\beta}'X'y + \hat{\beta}'X'X\hat{\beta}, && (1) \end{aligned}$$

where we used that the transpose of a scalar equals itself, so $(y'X\hat{\beta})' = \hat{\beta}'X'y$.

Differentiating the RSS

To find the $\hat{\beta}$ that minimizes $e'e$, differentiate (4) with respect to $\hat{\beta}$:

$$\frac{\partial(e'e)}{\partial\hat{\beta}} = -2X'y + 2X'X\hat{\beta} = 0. \tag{5}$$

To check this is a minimum, differentiate again with respect to $\hat{\beta}$:

$$\frac{\partial^2(e'e)}{\partial\hat{\beta}\,\partial\hat{\beta}'} = 2\,X'X,$$

which is positive definite when X has full column rank.

Normal Equations

From (5) we obtain the *normal equations*:

$$(X'X)\,\hat{\beta} \;=\; X'y. \tag{10}$$

Two immediate facts about $X'X$:

- It is always square ($k \times k$).
- It is symmetric.

Closed-Form Estimator

Since $X'X$ and $X'y$ are known from the data while $\hat{\beta}$ is unknown, if $(X'X)^{-1}$ exists, premultiply both sides of (10) by the inverse:

$$(X'X)^{-1}(X'X)\,\hat{\beta} \;=\; (X'X)^{-1}X'y. \tag{11}$$

Because $(X'X)^{-1}(X'X) = I_k$, this gives

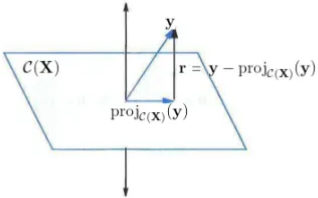
$$\hat{\beta} \;=\; (X'X)^{-1}X'y. \tag{12}$$

Up to this point, no stochastic assumptions were required. The OLS estimators are linear functions of the observed data (X, y). If X is random, we condition on its realization.

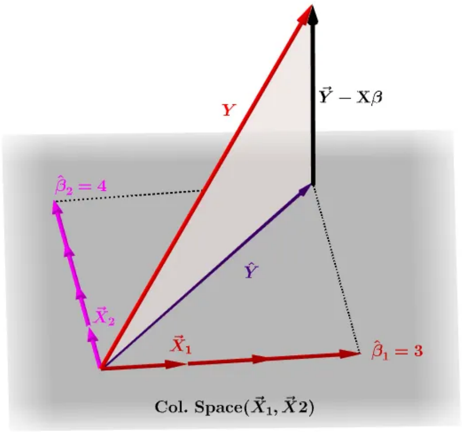
- **Full column rank:** $\mathrm{rank}(X) = k$ with $k < n$. Then $X'X$ is invertible and the normal equations have a unique solution.

Geometric Approach to OLS

Minimizing $\|\mathbf{r}\|^2 = \|\mathbf{y} - \mathbf{X}\beta\|^2 = \left\| \mathbf{y} - \left(\underbrace{\beta_0 \mathbf{x}_1 + \beta_1 \mathbf{x}_2 + \cdots + \beta_p \mathbf{x}_{p+1}}_{\text{linear combination of columns of } \mathbf{X}} \right) \right\|^2$ is equivalent to solving the equation $\mathbf{X}\hat{\beta} = \text{proj}_{\mathcal{C}(\mathbf{X})}(\mathbf{y})$ which represents the *orthogonal projection* of $\mathbf{y}$ on to the column space of the design matrix $\mathcal{C}(\mathbf{X})$ (set of all possible linear combinations of the columns of $\mathbf{X}$):



Geometric Approach to OLS: Data Space Visualization



Appendix

Matrix Derivative Rules Used

Let a, b be $K \times 1$ vectors and A a symmetric $K \times K$ matrix. We use:

$$\frac{\partial\, a'b}{\partial b} = a, \qquad \frac{\partial\, b'a}{\partial b} = a, \qquad \frac{\partial\, b'Ab}{\partial b} = 2Ab.$$

In our OLS derivation:

$$\frac{\partial\, \hat{\beta}'X'y}{\partial \hat{\beta}} = X'y, \qquad \frac{\partial\, \big(-2\,\hat{\beta}'X'y\big)}{\partial \hat{\beta}} = -2X'y, \qquad \frac{\partial\, \hat{\beta}'X'X\hat{\beta}}{\partial \hat{\beta}} = 2X'X\,\hat{\beta}.$$

These identities produce the gradient $\nabla_{\hat{\beta}}(e'e) = -2X'y + 2X'X\hat{\beta}$.

Linear Independence

Let $\{x_1, \ldots, x_r\}$ be $n \times 1$ column vectors. **Definition:** $\{x_1, \ldots, x_r\}$ is *linearly independent* iff

$$\alpha_1 x_1 + \cdots + \alpha_r x_r = 0 \;\Rightarrow\; \alpha_1 = \cdots = \alpha_r = 0.$$

Equivalent: no vector in the set can be written as a linear combination of the others.

Example. $\begin{bmatrix} 1 & 3 \\ 2 & 6 \\ 0 & 0 \end{bmatrix}$ has dependent columns (second is $3\times$ the first).

Rank of a Matrix

For $A \in \mathbb{R}^{n\times m}$ with columns $a_1, \ldots, a_m$:

$$\mathrm{rank}(A) = \max\{\text{number of linearly independent columns of } A\} = \dim \mathcal{C}(A).$$

Basic facts:

- $\mathrm{rank}(A) \leq \min(n, m)$, and $\mathrm{rank}(A) = \mathrm{rank}(A')$.
- $\mathrm{rank}(A) = m \Rightarrow A$ has *full column rank*.
- If A is $k \times k$ and $\mathrm{rank}(A) = k$, then A is invertible.

Why Rank Matters for OLS

Let $X = [x_1 \ \cdots \ x_k] \in \mathbb{R}^{n \times k}$.

- $\operatorname{rank}(X) = k$ (columns independent) $\Rightarrow X'X$ is $k \times k$ and invertible.
- Reason: for any nonzero z, $z'X'Xz = \|Xz\|^2 > 0 \Rightarrow X'X$ is positive definite.
- Then the normal equations have a unique solution:

$$(X'X)\hat{\beta} = X'y \quad \Rightarrow \quad \hat{\beta} = (X'X)^{-1}X'y.$$

- If $\operatorname{rank}(X) < k$: perfect multicollinearity $\Rightarrow X'X$ singular, coefficients not identified.